

Correspondence: Computing Geometry for Attrs

Source feature: LayoutData
Target feature(s): refX, refY, refHeight, refWidth
Automation type: OperatorHelp - MIX
Relation: Although GMF LayoutData and JointJS refX/refY/refWidth/refHeight are grounded in different layout paradigms, both mechanisms ultimately serve to encode the spatial intent of sub-figures within a container. GMF expresses this intent through nested XY offsets that are resolved imperatively during rendering, whereas JointJS captures the same intent declaratively by anchoring sub-elements relative to their parent's dimensions.

Operators: Mr. X

Main Strategy:

- 1. [A] check if LayoutData is a kind of XYLayoutData (X)
- 2. [-] if X is true
 - 2.1. [A] take XYLayoutData (A)
 - 2.2. [A] take the parent (B) of A
 - 2.3. [A] check if B's most top parent shape has a kind of size (Y)
 - 2.4. [-] if Y is true
 - 2.4.1. [A] take the size (preferred/min/max in order) of the most top parent shape (C)
 - 2.4.2. [A] take Dx (D) and Dy (E) of A
 - 2.4.3. [A] take Dx (F) and Dy (G) of C
 - 2.4.4. [A] take Zx=sum of all Xs of the XYLayoutDatas from A to C (H)
 - 2.4.5. [A] take Zy=sum of all Ys of the XYLayoutDatas from A to C (I)
 - 2.4.6. [A] calc: refWidth= D/F, refHeight= E/G, refX= H/F, refY= I/G
 - 2.4.7. [A] take the descriptor property (J) of B
 - 2.4.8. [A] check if there is a compartment\diagram label (K) that its figure property refers to J and its accessor property refers to a child access that its figure property refers to B (Z)
 - 2.4.9. [-] if Z is true
 - 2.4.9.1. [A] take the compartment\feature label mapping (L) that refers to K
 - 2.4.9.2. [A] take the equivalent object (M) of L in attrs
 - 2.4.9.3. [A] paste refWidth, refHeight, refx, and refY into M
 - 2.4.9.4. [M] operators should adjust the numbers if there are gaps or needs fine-tuning
 - 2.4.10. [A] else check if there is a node (N) that its figure property refers to the top most parent (W)
 - 2.4.11. [-] if W is true
 - 2.4.11.1. [A] take the node mappings (O) that refers to N
 - 2.4.11.2. [-] for every node mapping (P) in O
 - 2.4.11.2.1. [A] take P's Dia Element equivalent (Q)
 - 2.4.11.2.2. [A] if not exist, add an object (R) to attrs of Visual Attributes of Q with the name being the name property of B
 - 2.4.11.2.3. [A] paste refWidth, refHeight, refx, and refY into R
 - 2.4.11.2.4. [M] operators should adjust the numbers if there are gaps or needs fine-tuning

Structural overview

